LAB 4 — CONNECTIVITY OVERVIEW

Front and rear are the terminated ends. Cockpit taps both backbones.
Each zone has its own two isolation groups; termination remains on the bus side.
Input fuse is not per-motor current limiting. See Fuse-and-scope-review.md.

Front zone

+3V3
+6V6_SERVO
GND
CAN_A_H
CAN_A_L
CAN_B_H
CAN_B_L
+5V
VBAT_PROTECTED

+3V3
+6V6_SERVO
GND
CAN_A_H
CAN_A_L
CAN_B_H
CAN_B_L
+5V
VBAT_PROTECTED

File: Front.kicad_sch

Cockpit zone

+3V3
+5V
GND
CAN_A_H
CAN_A_L
CAN_B_H
CAN_B_L

+3V3
+5V
GND
CAN_A_H
CAN_A_L
CAN_B_H
CAN_B_L

File: Cockpit.kicad_sch

Rear zone

+3V3
+5V
GND
CAN_A_H
CAN_A_L
CAN_B_H
CAN_B_L
VBAT_PROTECTED

+3V3
+5V
GND
CAN_A_H
CAN_A_L
CAN_B_H
CAN_B_L
VBAT_PROTECTED

File: Rear.kicad_sch

Power

+3V3
+5V
+6V6_SERVO
GND
VBAT_PROTECTED

+3V3
+5V
+6V6_SERVO
GND
VBAT_PROTECTED

File: Power.kicad_sch

CAN_A_H/L and CAN_B_H/L are separate root nets. Same-name local labels on this root join the shown sheet pins. Child labels connect only through those ports.

Power flags document external battery power and outputs after passive filters; they do not prove current capacity, regulation, thermal behavior or PCB completeness.

See November2-Anson-review.md for review changes and verification.

Shared nets: two CAN backbones, power and ground only.
Electrical checks do not replace prototype testing.

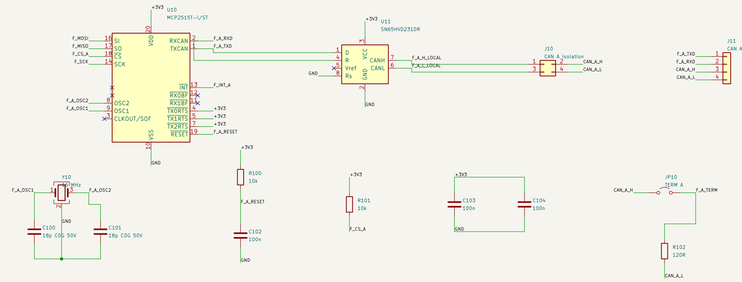
Team 4

Sheet: /
File: Lab4.kicad_sch

Title: Lab 4 — three-zone controller

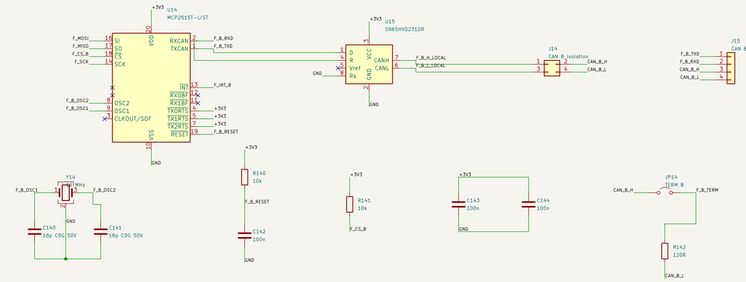
Size: A2	Date: 2026-11-02	**Rev: Review 2**
KiCad E.D.A. 10.0.4		Id: 1/5

CAN A: controller, transceiver and isolation



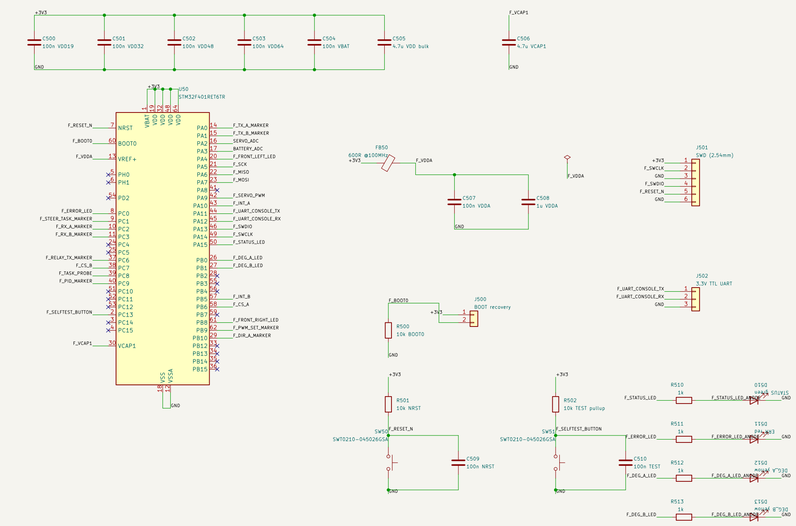
16 MHz: SPI from zone processor

CAN B: controller, transceiver and isolation



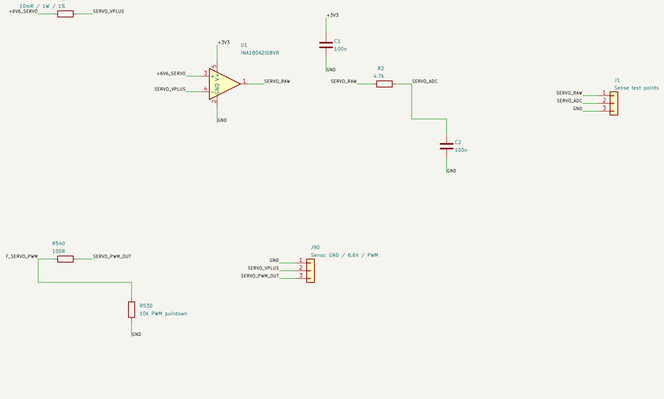
16 MHz: SPI from zone processor

STM32F401: processor, supplies and recovery

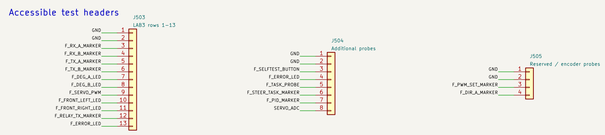


100 nF at each VDD pin; VCAP is a dedicated regulator output.

Servo supply, high-side current sense and PWM

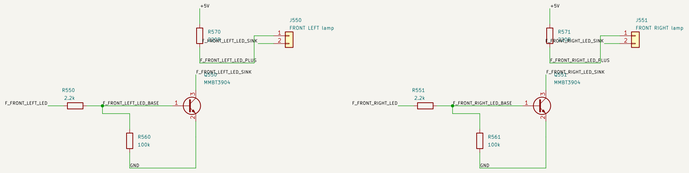


Accessible test headers

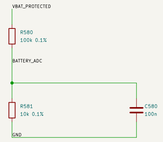


Lab 3 pin assignments retained. Do not apply 5 V to MCU/ADC inputs.

Vehicle lamps



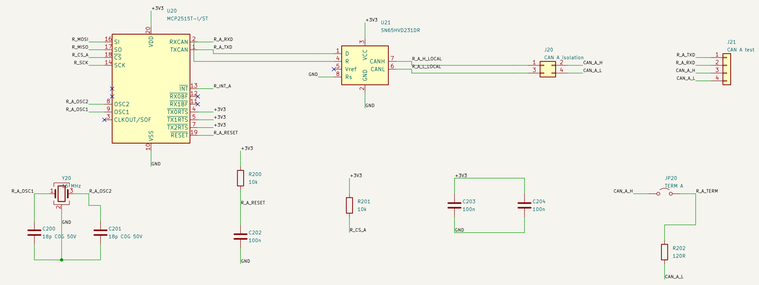
Battery voltage monitor



+3V3 +6V6_SERVO GND CAN_A_H CAN_A_L CAN_B_H CAN_B_L +5V VBAT_PROTECTED

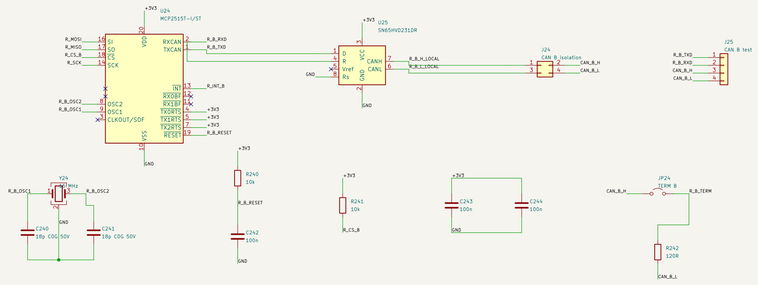
Team 4	
Sheet: /Front zone/	
File: Front.kicad_sch	
Title: Front: steering, telemetry and dual CAN	
Size: A1	Date: 2026-11-02
KiCad E.D.A. 10.0.4	**Rev: Review 2**
	Id: 2/5

CAN A: controller, transceiver and isolation



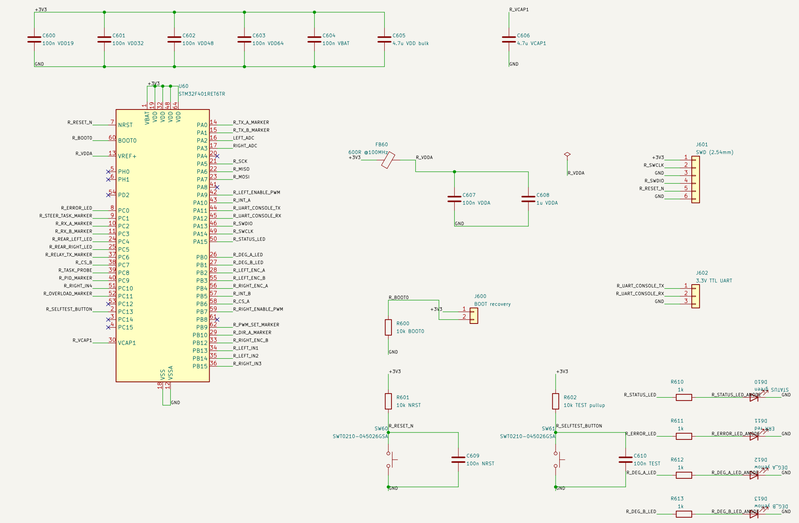
16 MHz; SPI from zone processor

CAN B: controller, transceiver and isolation



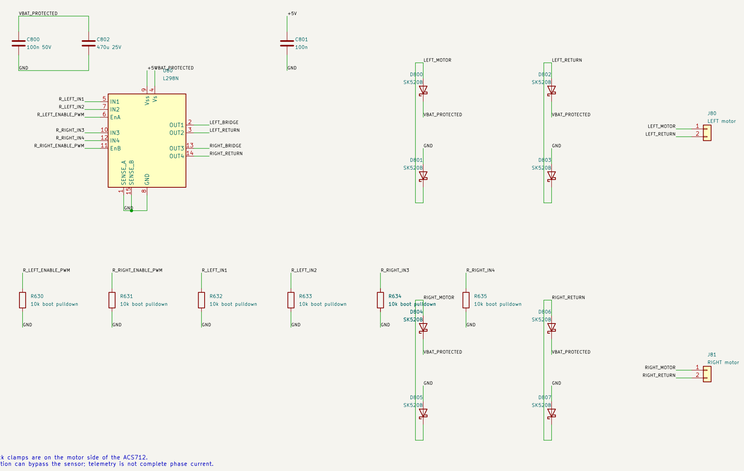
16 MHz; SPI from zone processor

STM32F401: processor, supplies and recovery



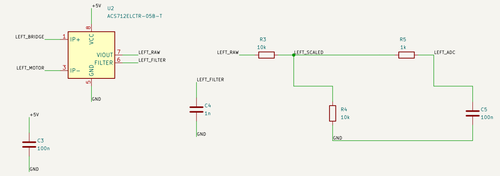
100 nF at each VDD pin; VCAP is a dedicated regulator output.

Motor H-bridge / output clamps / connectors

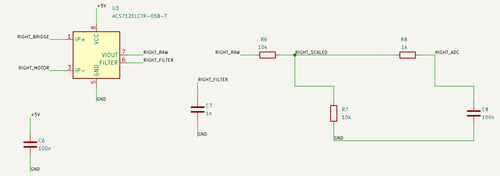


All flyback clamps are on the motor side of the ACS712.
Recirculation can bypass the sensor; telemetry is not complete phase current.

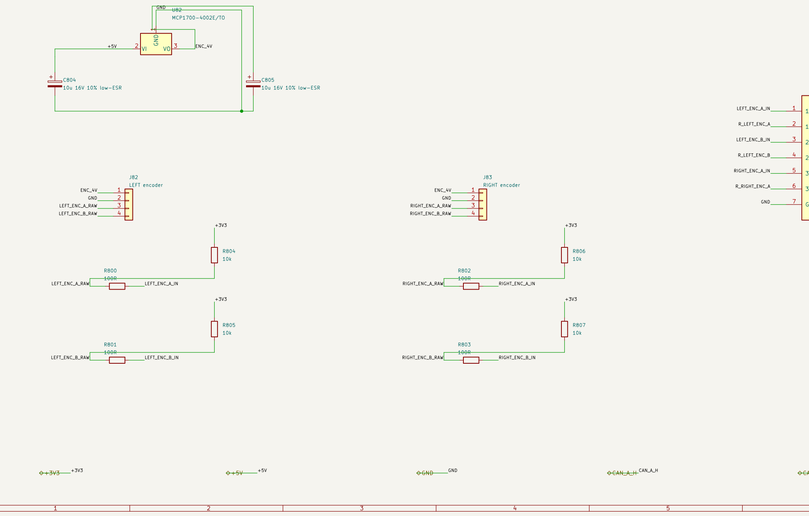
Left motor current / ADC filter



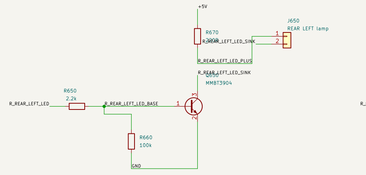
Right motor current / ADC filter

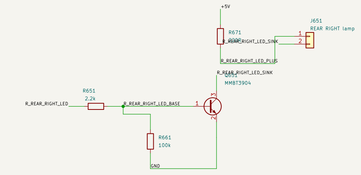


Encoder power / 5 V-tolerant Schmitt inputs

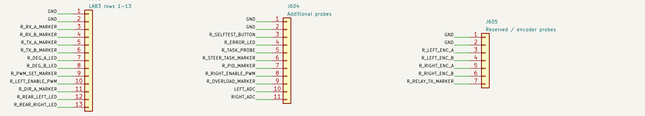


Vehicle lamps

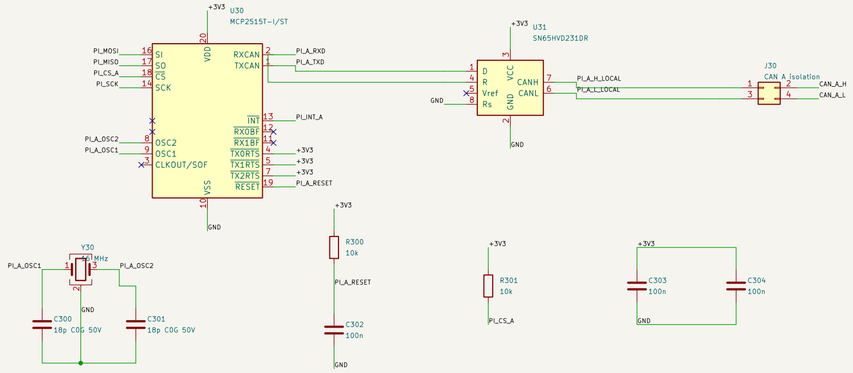




Accessible test headers

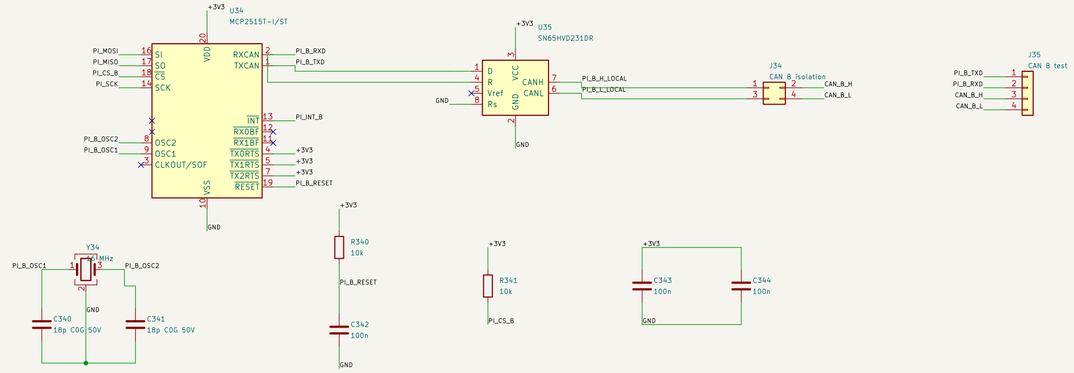


CAN A: controller, transceiver and isolation



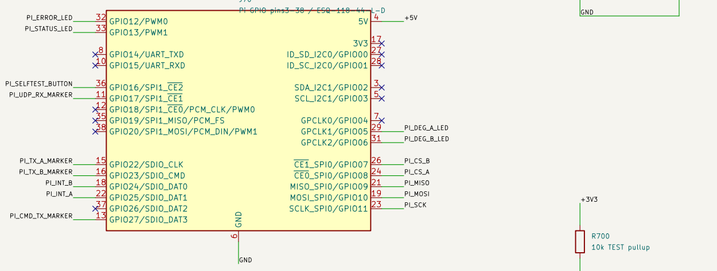
16 MHz: SPI from zone processor

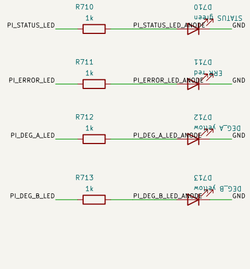
CAN B: controller, transceiver and isolation

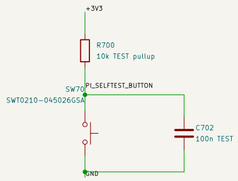


16 MHz: SPI from zone processor

Pi interface: physical header pins 3–38







Accessible test headers



J70 uses a 36-contact socket aligned with Pi GPIO pins 3–38.
Pin 2 is supplied through Pi pin 4; pins 1/39/40 are outside this socket.
Power and ground connect to the PCB; other used pins carry 3.3 V SPI/GPIO.
The Pi has no external USB accessories in the design load case.

CAN termination is at the front and rear zones only.
Remove both isolation shunts to disconnect one local transceiver.

+3V3 +3V3 +5V +5V GND GND CAN_A_H CAN_A_H CAN_A_L CAN_A_L CAN_B_H CAN_B_H CAN_B_L CAN_B_L

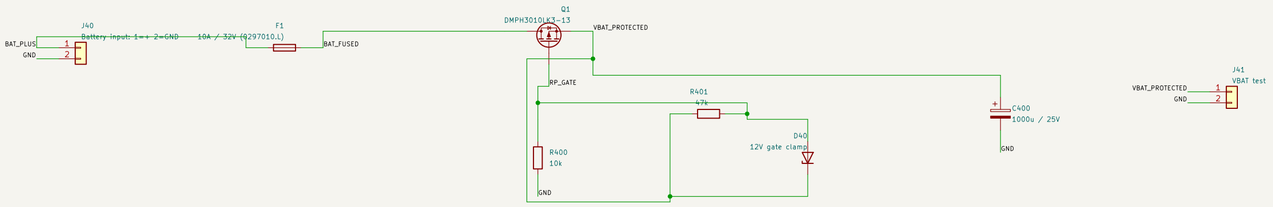
Team 4

Sheet: /Cockpit zone/
File: Cockpit.kicad_sch

Title: Cockpit: Raspberry Pi interface and dual CAN

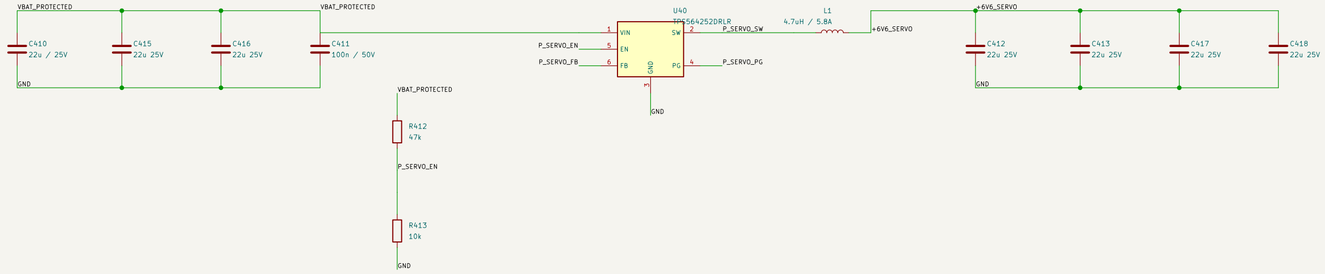
Size: A2	Date: 2026-11-02	**Rev: Review 2**
KiCad E.D.A. 10.0.4		Id: 4/5

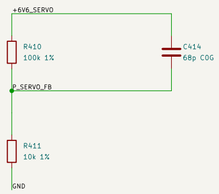
3S LiPo input / fuse / reverse-polarity protection



F1: selected 10 A MINI fuse; holder 3568. Input-path fuse, not per-motor current limiting.
Verify the installed fuse and harness before battery operation; see Fuse-and-scope-review.md.

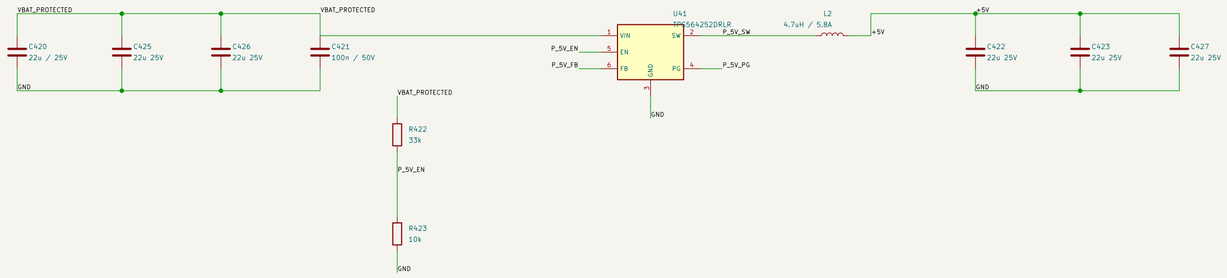
Servo rail: 6.6 V

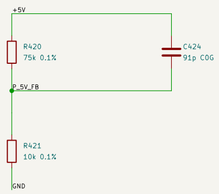






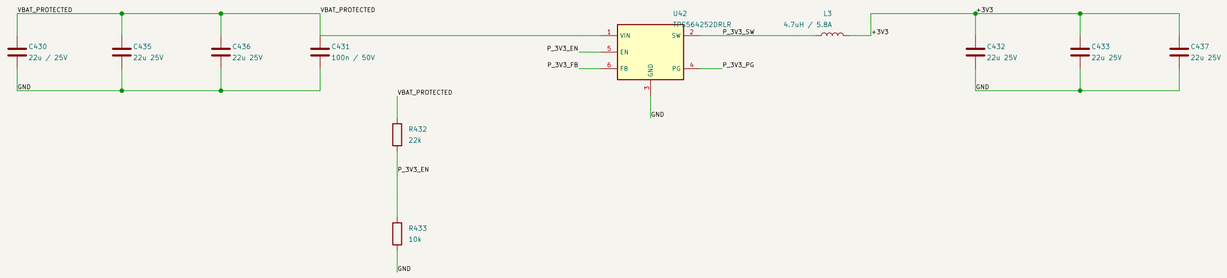
Pi / L298 logic rail: 5.1 V

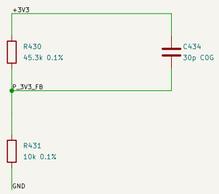






Logic rail: 3.3 V







BAT_PLUS GND VBAT_PROTECTED +3V3 +5V +6V6_SERVO

+3V3 +3V3 +5V +5V GND GND +6V6_SERVO +6V6_SERVO VBAT_PROTECTED VBAT_PROTECTED

Team 4	
Sheet: /Power/	
File: Power.kicad_sch	
Title: Power: input protection and regulated rails	
Size: A2	Date: 2026-11-02
KiCad E.D.A. 10.0.4	**Rev: Review 2**
	Id: 5/5